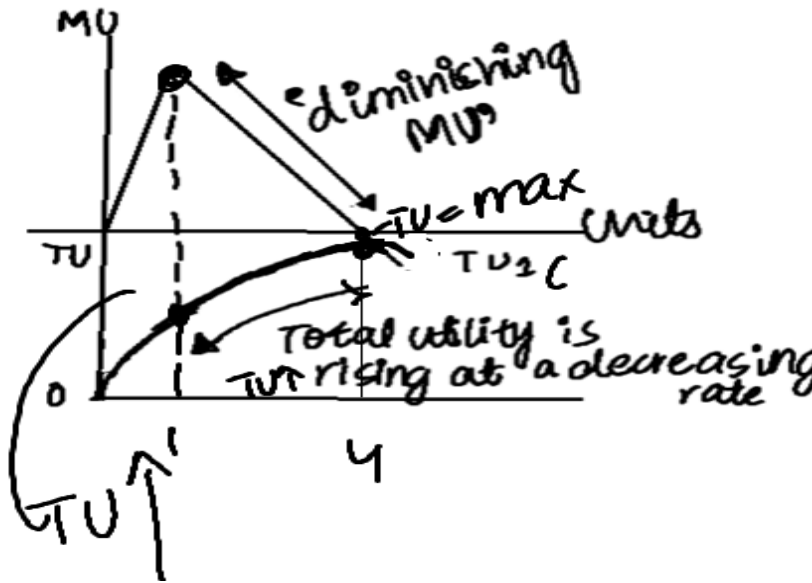


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1-C:

- ➔ When Marginal utility is positive but is falling, this represents the case of “diminishing marginal utility”. When marginal utility (additions to the TU upon consumption of an additional unit) diminishes, Total utility increases “but at a decreasing rate”.

This can be seen in the diagram below:



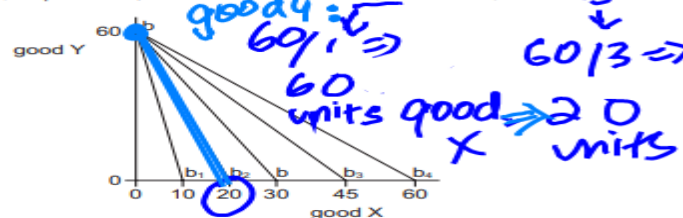
Additionally, as can be seen in option D, total utility increases at an increasing rate only when Marginal utility increases. Further, total utility falls when Marginal utility is negative, which explains why all other options will be incorrect.

2-B:

- ➔ When the price of Y is \$1, given the fixed income of \$60 per week, the maximum quantity of good Y that can be purchased will be: 60 (i.e. $60/1$)
- ➔ Additionally, when the price of good X has risen to \$3, given the fixed income of \$60 per week, the maximum quantity of good X that can be purchased will be : 20 (i.e. $60/3$)

That being said, key B, will be correct **because the new budget line will be “b-b2”**

A budget line (b-b) shows the combinations of two goods, X and Y, that a consumer can obtain within a fixed income of \$60 per week, when the price of Y = \$1 and X = \$2.



If the price of X rises to \$3, what will be the new budget line?

- A b-b₁ **B b-b₂** C b-b₃ D b-b₄

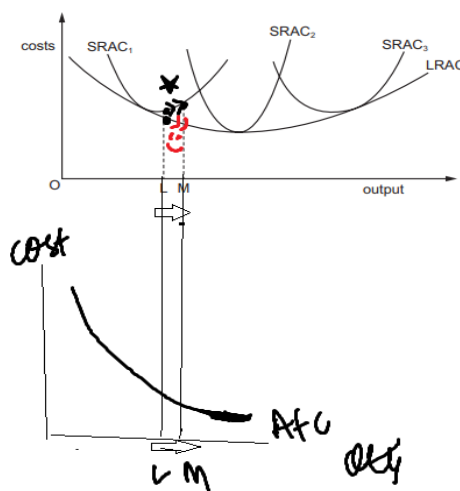
3-D:

A contestable market is one where entry and exit barriers are low, making it easy for firms to enter or leave the market. This increases the potential threat of competition, even when the number of firms that exist in the market currently is less.

- Perfect competition: Yes, contestable. Perfect competition assumes free entry and exit, making it a contestable market.
- Monopolistic competition: Yes, contestable. Monopolistic competition has very less barriers to entry and exit, making it contestable.
- Pure monopoly: No, not contestable. A pure monopoly has a single firm with significant barriers to entry, making it non-contestable.

Note: A monopoly does face a contestable situation (that is when a monopoly being a monopoly cannot act as a monopoly due to the potential threat of competition) but a pure monopoly does not face a contestable situation.

4-D:



* upward movement along the SRAC curve $\rightarrow$ law of diminishing returns
c downward movement along the LRAC curve $\rightarrow$ economies of scale

A – Average fixed cost keeps on falling as output is increased. This is why, key A is a valid statement as in this question, output increases from L to M.

B – Since there is a downward movement along the Long run Average cost, this represents that there are economies of scale that are being experienced.

C – When we increase output from L to M, there will be an upward movement along the short-run Average cost curve, which indicates that law of diminishing returns will be operative.

D – However, as discussed above in explanation of key B, it can be said that increasing returns are experienced. **Since there are no dis-economies of scale being experienced, this option is correct since the statement mentioned is invalid.**

5-C: In an oligopolistic market, firms are interdependent, and their actions affect each other. When an oligopolist changes its price, rivals may react in various ways, leading to a kinked demand curve. Here's why option C is correct:

Assume an oligopolist (firm A) increases its price:

- Rivals (firm B, C, etc.) might ignore the price rise, as they don't want to lose market share by following suit. This means they won't increase their prices.
- Firm A's demand curve will be relatively elastic (flat) above the initial price because **this represents the firm's action**, as customers can easily switch to rivals' products.

Now, assume firm A decreases its price:

- Rivals will likely copy the price fall to maintain their market share and avoid losing customers.
- Firm A's demand curve will be relatively inelastic (steep) because this represents the **industry's action**.

6-A:

Price discrimination is a pricing strategy that involves charging different prices to different customers for the same product or service. A key condition for effective price discrimination is that the product or service cannot be resold or transferred from one customer to another. This is known as the "no resale condition"

If the product can be resold, then customers who are charged a lower price can resell it to those who are charged a higher price, reducing the effectiveness of price discrimination.

Here's why the other options are incorrect:

B) Price inelasticity is not a necessary condition for price discrimination. **Instead, varying elasticities are required to discriminate among customers with different elasticities, as consumers with low PED are typically charged high prices, and those with high PED are charged low prices.**

C) Subsidizing one market with profits from another is not a condition for price discrimination.

D) Low barriers to entry are not a condition for price discrimination. In fact, price discrimination is often more feasible when there are high barriers to entry, as it reduces the risk of competition.

7-A:

- High fixed costs: New firms must invest heavily in assets, equipment, and infrastructure, making it difficult to enter the industry.
- High MES: The industry's optimal production scale is large, requiring significant investment and making it hard for new firms to achieve efficiency.
- High sunk costs: Investments are irreversible, and assets cannot be easily sold or transferred, making it costly for new firms to exit if they fail.

This combination creates significant entry barriers, making it challenging for new firms to enter the industry.

8-B:

Dynamic efficiency refers to the ability of a firm or industry to reduce its long-run average cost curve over time, typically through technological progress or innovation because of re-investment of super-normal profits. This is represented graphically by a downward shift in the long-run average cost (LRAC) curve, as a result of which, average costs at the given level of output decreases.

Here's what the other options represent:

A) A downward move to the minimum point of a long-run average cost curve represents a firm moving to the optimal point on its existing LRAC curve, which **indicates productive efficiency**.

C) A move from a point inside a production possibility frontier (PPF) to a point on it represents a move from inefficient to efficient production, which indicates **pareto improvement**.

D) A reduction in marginal revenue so that it equals long-run marginal cost represents a firm is equating $MC=MR$, which indicates that **it is operating at the profit maximizing level of output**.

9-D:

The correct answer is D) the cost to a fishing fleet of relocating to another port.

External costs, also known as negative externalities, are costs that fall on parties outside of the main transaction. The cost to a fishing fleet of relocating to another port is an external cost because the fishing fleet is not directly involved in the decision, but may be affected by the terminal's construction (e.g., due to increased traffic or loss of docking space).

A) The cost of a ticket for a passenger to travel on a cruise ship is a private cost, borne by the individual passenger.

B) The cost of fuel to operate the cruise ships is a private cost, borne by the cruise ship operators.

C) The costs paid by cruise ships when they dock at the port are also private costs, borne by the cruise ship operators.

10-B:

Private costs are the costs that a company directly pays, like labor, materials, and transportation. These are costs that are reflected in the price of the product or service.

Additionally, since now the company is also required to pay for external costs created by its actions, an example of which is environmental damage, it can be said that it has to pay for external costs too now.

Therefore, the total costs will be $PC + EC$.

11-C:

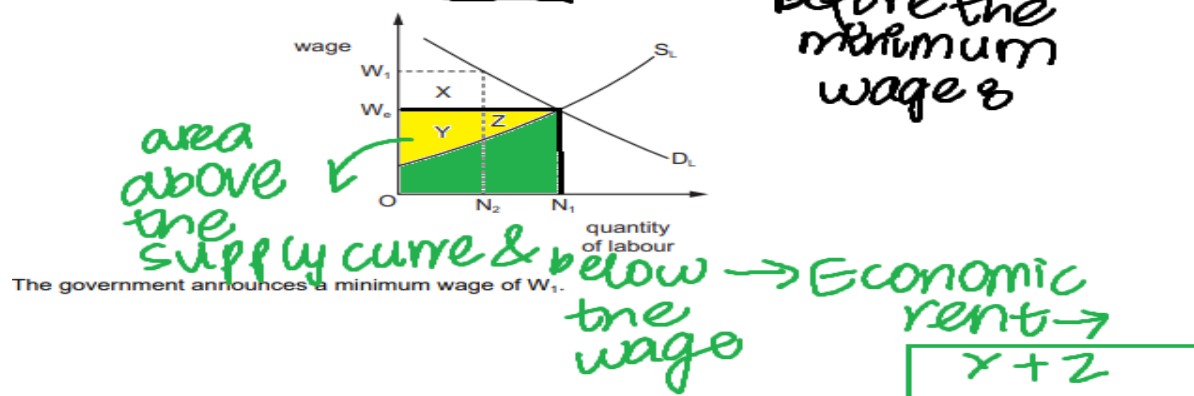
Government failure refers to situations where government intervention in the economy leads to inefficient or undesirable outcomes. Options A, B, and D are all examples of government failure:

- A) Inefficient allocation of resources
- B) Net loss of economic welfare
- D) Social costs exceeding social benefits

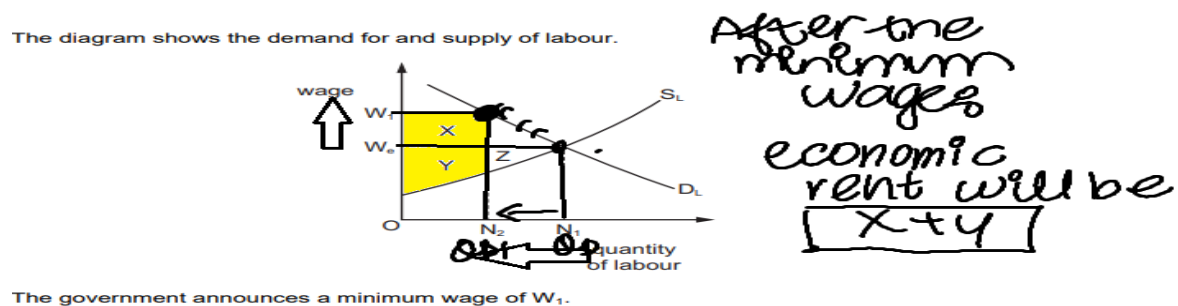
On the other hand, an increase in consumer surplus (option C) is actually a desirable outcome, indicating that consumers are better off due to government intervention. This could be achieved through policies like consumer protection or regulation that promotes competition.

12-A:

The diagram shows the demand for and supply of labour.



The diagram shows the demand for and supply of labour.



As can be seen above, **it can be said that, economic rent after the minimum wage legislation rises by X.**

13-D:

The correct answer is D) a shift of the supply curve for its labour to the right.

By introducing its own training scheme, the firm is investing in human capital, which will increase the supply of skilled labor to the firm. This is likely to shift the supply curve for labor to the right, indicating an increase in the quantity of labor supplied at each wage level.

Here's why the other options are incorrect:

A) The firm is trying to address a labor shortage, so it's unlikely to decrease the number of workers it employs.

B) The training scheme is likely to increase productivity, not decrease it, as the firm is investing in the skills and development of its workforce.

C) Geographical mobility refers to the ability of workers to move to different locations for work. The training scheme is unlikely to directly affect geographical mobility. **Instead, if workers are retrained, their occupational mobility will improve.**

14-D:

A regressive tax is a tax that takes a larger percentage of income from low-income individuals than from high-income individuals. Introducing a new regressive tax will likely increase tax revenues for the government, as it will impose a greater burden on low-income individuals who may have not contributed much to the tax revenues with progressive taxes. **This explains why key D will be correct.**

A and B) Aggregate demand may actually decrease, as when the prices of goods and services rise, the real incomes fall, which subsequently translates into reduced consumption expenditure and in turn Aggregate demand falls too. **When aggregate demand falls, production and subsequently demand for labor falls that results in increased demand deficient unemployment.**

C) Income equality will likely worsen, as the tax burden falls disproportionately on low-income individuals, increasing the income gap between the rich and the poor.

15-A:

Option A)

Advantage: targets those in need (food subsidies directly benefit low-income individuals and families who need assistance the most)

Disadvantage: high administrative costs (implementing and managing food subsidy programs can be complex and costly)

B) Higher national minimum wage: It does not encourage skill training. Additionally, it is also unlikely to create a budget deficit.

C) Provision of free child care: Employability refers to the ability of a person to gain and maintain employment, and to move between jobs and careers throughout their life. Free childcare means that parents, especially mothers, don't have to worry about the financial burden of childcare, making it easier for them to enter the workforce or increase their working hours. **This is how, it will increase employability.**

However, it may not result in dis-incentives to work, as this will encourage parents to take up work.

D) Universal monetary benefits can save on bureaucracy because they often have straightforward eligibility criteria, such as age or citizenship, reducing the complexity of means-testing and eligibility assessments. However, they may not reduce labor mobility.

16-D:

> Higher output is required, and to achieve that, subsidies are given, this means that the good is a merit good, which generates “external benefits” or “positive externalities”

> If the demand for merit goods, for example, education is “elastic”(>1), this means that when subsidies increase supply, and in turn result in a “fall in price”, quantity demanded will extend by a relatively higher percentage, which will be the “most beneficial outcome”, as there will be a greater likelihood of output increasing to socially optimum level of output. This explains why, key D, will be correct.

17-D:

A – When an economy was closed, the equation of AE was $C+I+G$. Now that it is open or open to international trade, another variable i.e. net exports will be included, which will make the equation “ $C+I+G+(X-M)$ ” now.

B – Since export revenues now occur, the number of injections into the circular flow will increase.

C – Since import expenditures now occur, the number of leakages from the circular flow will decrease.

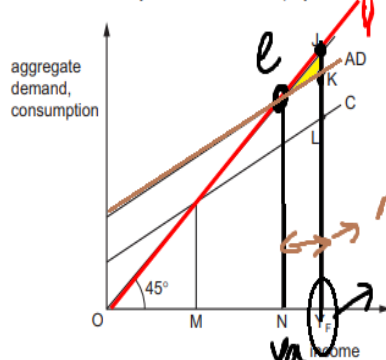
D - The multiplier effect is a key concept in macroeconomics, representing how an increase in spending or investment can have a multiplied effect on economic activity. However, when an economy opens up to international trade, the multiplier effect can be reduced due to the increased leakages from **imports**.

Note: An increase in marginal propensity to import represents increased leakages, which will reduce marginal propensity to consume, thereby reducing the value of the multiplier.

18-A:

Deflationary gap is measured by seeing how much does Aggregate expenditure fall short of real national income at the full employment level of output.

The diagram shows a closed economy in which the full employment level of income is Y_F .



$Y = I + C$
 $AD = C$
 $AD < Y$; deflationary gap of "JK"

negative output gap of NY_F
 full employment level of income

Which distance measures the deflationary gap?

- ☒ A JK ☐ B JL ☐ C NY_F ☐ D MY_F

-ve output gap

19-D:

Key take-away: **A regulation is a rule/ law/ limit or a ban.**

The setting of industrial emission standards is considered regulatory rather than market-based because it:

- Sets specific limits on emissions.

The other options are market-based policies:

A) Carbon taxes: Places a price on pollution to reduce emissions.

B) Tradable pollution permits: Creates a market for emissions allowances, allowing companies to buy and sell permits.

C) Government subsidies: Provides financial incentives to encourage reduced emissions or clean energy investments.

20-B:

This cause of economic growth is most likely to ensure sustainable growth because:

- Environmental education promotes awareness and adoption of sustainable practices
- Taxes on harmful goods create an economic incentive to reduce pollution and resource depletion
- This approach balances economic growth with sustainability, as a result of which, **option B becomes correct.**

The other options may actually hinder sustainable growth:

A) Deregulation of industries using mineral resources and fossil fuels can lead to overexploitation and environmental degradation.

C) Trade barriers can limit access to eco-friendly technologies and ideas from abroad.

D) Reduced planning regulations in areas of natural beauty can lead to environmental destruction and unsustainable development.

21-B:

Frictional unemployment exists in any economy due to people being in the process of moving from one job to another.

One important type of frictional unemployment is “search unemployment” that occurs when workers are searching for jobs and it is a result of imperfect information about job vacancies and matching frictions in the labor market.

Therefore, by opening centers where job vacancies are advertised, the government is improving labor market information and reducing search costs for workers, thereby reducing frictional unemployment. This explains why key B is correct.

A) Cyclical unemployment:

- Cyclical unemployment is a type of unemployment that occurs when the economy is in a recession
- It is caused by a decrease in aggregate demand, leading to a reduction in production and demand for labor which subsequently translates into un-employment.
- To address cyclical unemployment, the government would need to implement policies that stimulate economic growth, such as **expansionary (reflationary) fiscal or monetary policies.**

C) Regional unemployment

- It is categorized as a type of structural unemployment_ that occurs when there are changes in the structure of the economy, mainly pattern of demand and supply.
- When unemployment is concentrated in a particular area due to changes in structural factors, this is regarded as **regional unemployment.**

Example:-Mpumalanga province in South Africa was once a major coal mining hub, but the industry has been declining in recent years due to the demand switching to more environmentally friendly sources of renewable energy. This decline has led to significant job losses and regional unemployment in the area, as many communities were heavily reliant on the coal mining industry.

To address regional unemployment, the government would need to implement policies that promote regional development, such as investing in infrastructure or providing incentives for businesses to locate in disadvantaged regions.

D) Seasonal unemployment:

- Seasonal unemployment occurs when industries have fluctuating demand for labor due to seasonal changes.
- Examples include industries such as agriculture, tourism, and construction.
- To address seasonal unemployment, the government would need to implement policies that provide support to industries during their off-seasons, such as training programs or subsidies to maintain employment levels.

22-C:

The Accelerator Theory posits that investment is driven by changes in national income, rather than the level of national income. The theory suggests that businesses invest more in capital goods when national income is growing rapidly, as this indicates an increasing demand for goods and services. Conversely, when national income growth slows down, investment decreases.

In other words, the Accelerator Theory assumes that investment is a function of the rate of change in national income (ΔY)

Therefore, the co-efficient of accelerator is calculated using the formula: **Change in Investment/ Change in income.**

Assumptions (/) Capital-output ratio 1 : 10

Case I. Life of the Machine 10 years.

	Period	Change in consumption	Capital Equip-ment needed	Gross Investment			Percentage change in Gross investment
				Additions	Replacement	Total	
10% rise in demand	0	1000	100	Nil	10	10	100% increase
	1	1100	110	10	10	20	

This can be seen in the table below.

- A) This option is incorrect because the Accelerator Theory focuses on the rate of change in national income, not business confidence.
- B) This option is incorrect because the Accelerator Theory does not consider savings as a direct determinant of investment.
- D) This option is incorrect because the Accelerator Theory does not consider interest rates as a direct determinant of investment. Instead, the marginal efficiency of investment theory describes a negative relationship between the rate of interest and investment.

23-A:

A) This option is correct because a depreciation of the exchange rate makes a country's exports cheaper and more competitive in the global market. As a result, there is an increase in demand for exports, and that too by a relatively higher percentage when the demand is elastic. When because of this, export revenues rise, there is an increase in Aggregate demand which leads to an increase in demand for labor, production and employment.

B) This option is incorrect because higher interest rates attract hot money inflows, but this does not lead to increased consumer spending. In fact, higher interest rates increase borrowing costs, which can lead to decreased consumer spending especially on **big ticket items**.

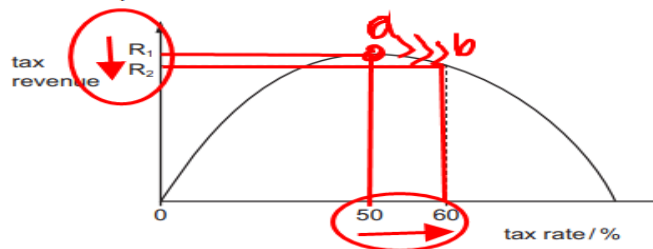
C) This option is incorrect because decreased government spending leads to fewer workers being employed, not more since it is a part of **deflationary fiscal policy**. Reduced government spending means fewer government contracts, fewer public sector jobs, and reduced economic activity. **When because of this, incomes reduce, this can lead to a decreased demand for imports, not increased demand.**

D) This option is incorrect because while a reduction in income taxes may increase consumer spending after having increased disposable incomes, it does not necessarily lead to increased demand for exports.

24-D:

The Laffer curve is a theoretical representation of the relationship between tax rates and revenue, and it suggests that there is an optimal tax rate that maximizes revenue. Beyond a certain point (in this case, 50%), tax revenue will decrease.

The Laffer curve shows expected tax revenues at different tax rates.



This is because, there is an increased incentive for tax evasion that is when individuals and firms find loopholes to evade tax (which explains why key B is incorrect). One way individuals can evade income tax is by spreading incomes into different accounts. Similarly, firms can choose to engage in cash transactions, to avoid being taxed.

Further, this brings with it, an increased incentive for emigration to lower tax economies (key A)

In addition, this decreases incentive for people to work (key C), or in other words, increases incentives for people to work fewer hours. This is because they may feel that the government is taking too large a share of their earnings. (key D)

25-A:

A) Strength: The means-tested benefits ensure that only those who need them most receive support, targeting the poorest segments of society (**as they are provided to low income workers**) and improving income distribution.

Weakness: The policy only assists **low-income workers** (only those who are working), leaving poor non-workers (e.g., those unable to work due to disability or caregivers) without any additional support, potentially exacerbating their poverty.

B) Intergenerational equity is not directly related to this policy which is based on equitable distribution of benefits among generations. However, because of an increase in tax rates on highest paid workers, tax evasion or tax avoidance is most likely to increase.

C) The policy may not encourage workers to work more hours, as the means-tested benefits might reduce the incentive to work additional hours. Additionally, because of this, the taxation system has become more progressive.

D) **Note: consider the efficiency versus equity argument here.**

While this policy may be equitable, this will not be efficient to operate. Additionally, this will affect income distribution more than wealth distribution.

26-A:

The table shows data on the unemployment and inflation rates for four countries in 2010 and 2014.

country	unemployment rate (% of workforce)		inflation rate (% per annum)	
	2010	2014	2010	2014
Australia	5.2	6.1	2.7	1.6
Canada	8.0	6.9	2.2	1.9
Japan	5.1	3.6	-0.3	2.5
United States	9.6	6.2	1.7	0.6

The Phillips Curve is a historical inverse relationship between unemployment rates and inflation rates. It suggests that as unemployment falls, inflation rises, and vice versa.

In the context of the question, options A (Australia and Japan) exhibit a typical Phillips Curve relationship.

This explains why only key A can be correct.

27-D:

To calculate a **Measure of Economic Welfare (MEW)** from **Gross Domestic Product (GDP)**, the following adjustments are made:

$$\rightarrow \text{MEW} = \text{GDP} + \text{value of leisure} + \text{value of unpaid volunteer work} - \text{environmental damage} - \text{crime rate}.$$

1. **Environmental costs:** GDP only considers the value of goods and services produced, without accounting for the environmental degradation or resource depletion that may have occurred in the process. **MEW subtracts environmental costs, such as pollution and resource depletion, to get a more accurate picture of economic welfare.**
2. **Value of leisure activity:** GDP only considers market transactions, ignoring the value of leisure time and non-market activities. **MEW adds the value of leisure activity to recognize the importance of free time and personal activities in contributing to overall well-being.**
3. **Value of unpaid work:** GDP only considers paid work, ignoring the value of unpaid work such as household chores, caregiving, and volunteering. **MEW adds the value of unpaid work to recognize the important contributions these activities make to economic welfare.**

28-B:

The correct answer is B) the quantity of a currency required to purchase the same basket of goods in both countries.

Purchasing Power Parity (PPP) is a method used to adjust for differences in the purchasing power of different currencies, allowing for more accurate comparisons of economic indicators across countries. PPP exchange rate is defined as the amount of currency required to purchase the same basket of goods and services in both countries.

A) This definition (**the amount of the domestic currency needed to purchase a unit of foreign currency**) describes the exchange rate, not PPP.

C) This definition describes the concept of "purchasing power" in general, but not PPP specifically.

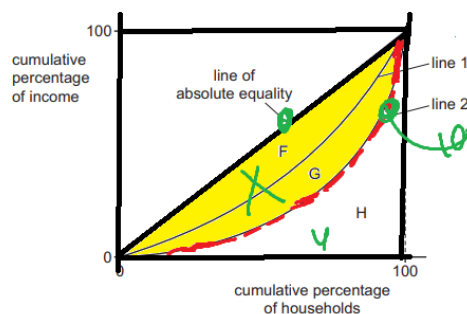
D) This definition describes the total value of household expenditure in a country, not PPP.

29-B:

➔ **Gini-coefficient** is calculated by dividing the area between the line of perfect equality and the Lorenz curve i.e. region X, with the total area under the curve i.e. region X and Y.

➔ **Gini-coefficient** -> $X/(X+Y)$

A Lorenz curve measures the income distribution in a country. The further the Lorenz curve is from the line of absolute equality, the greater the level of inequality.



gini coefficient = $\frac{X}{X+Y}$

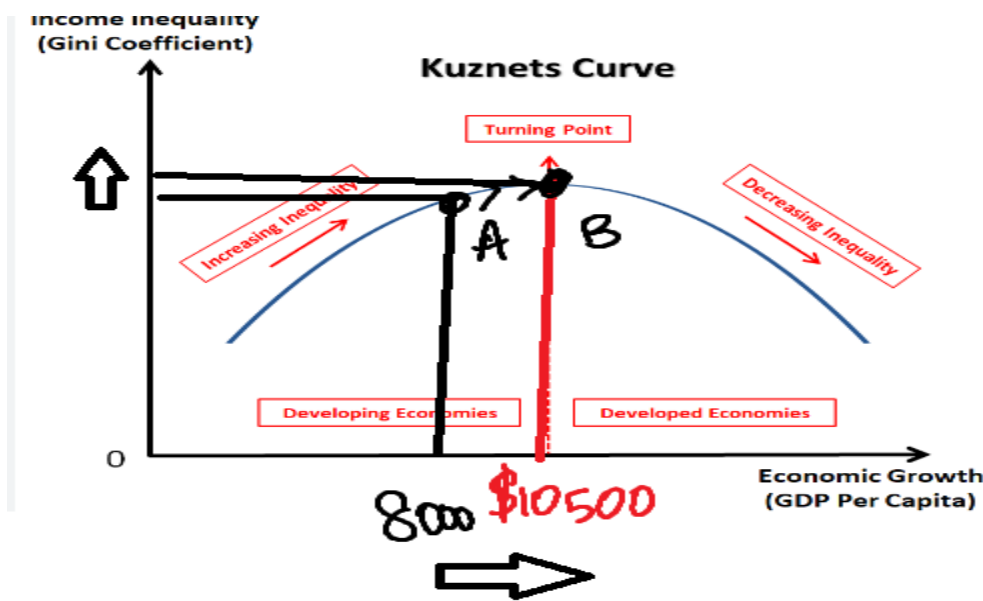
gini coefficient = $\frac{F+G}{F+G+H}$

What is the value of the Gini coefficient when the Lorenz curve is at line 2?

- A $\frac{F+G}{H}$ **B $\frac{F+G}{F+G+H}$** C $\frac{G}{G+H}$ D $\frac{H}{F+G+H}$

This explains why key B can be correct.

30-D:



The Kuznets curve suggests that as a country's GDP per capita increases, income inequality initially increases, but eventually decreases after a certain **turning point** (estimated to be around \$10,500 in this case).

Since the country's current GDP per capita is \$8,000, which is below the turning point, it is likely in the phase where income inequality increases with economic growth. This means that as the economy grows and GDP per capita increases, the wealthy tend to benefit more than the poor, leading to increased income inequality.

This explains why key D will be correct.

